GARL: Grasp-Aware Reinforcement Learning for Robust and Dexterous In-Hand Rotation

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Abstract: Dexterous multi-fingered hands on robotic arms are often subject to varying orientations and accelerations, yet most reinforcement learning (RL) methods for in-hand manipulation assume a fixed hand pose, limiting robustness and generalization. Our approach incorporates both object-related and hand-related considerations within a RL pipeline (Proximal Policy Optimization in our case) to achieve desired behaviors for dexterous manipulation tasks. It targets key aspects of multifingered grasps through quantitative measures of guaranteed stability, controllability, and manipulability. This enables a comprehensive analysis of the learned policies, characterizing the safety margin against undesired grasp configurations. It favors finger-gaiting sequences characterized by perturbation-invariant in-hand rotation with up to 12× fewer training steps and improves generalization to unseen perturbation intensities by 25×. The approach also enhances task performance by 18.1% and reduces average joint torques by 26.7%. Object generalization experiments demonstrate that our method provides consistent and reliable training outcomes. These simulation-only results bring the required robustness for future real-world deployment on humanoid robots.

1 INTRODUCTION

Robustness remains an understudied aspect of in-hand manipulation. In practical deployments, however, multi-fingered hands are typically mounted on robotic arms, where both the hand's orientation relative to gravity and the arm's accelerations vary continuously. Most prior works (Qi et al., 2022; OpenAI et al., 2019; Ma et al., 2023) assume a fixed hand orientation and, consequently, tend to perform poorly when evaluated under these unseen orientations. To achieve robustness, in-hand manipulation policies must actively regulate contact interactions to counter variations in external force direction and magnitude, preventing slippage or object loss while ensuring precise object motion.

To the best of our knowledge, only Yang et al. (Yang et al., 2024) have directly addressed this problem by training policies across multiple hand orientations. While this represents a significant advance, their approach still exhibits a limitation: a 42% performance drop between the most and least favorable orientations. This gap indicates that merely training on more orientations does not provide a fundamental solution. Achieving robust in-hand manipulation requires a principled approach that accounts for key aspects of multifingered grasps (Roa and Suárez, 2015; Prattichizzo and Trinkle, 2008). Instead of relying exclusively on extensive domain randomization, our method incorporates an inductive bias derived from analytical grasp formulations.

Configurations unsuitable for robust in-hand manipulation (e.g., missing contacts, uneven contact distribution, kinematic singularities, or closely spaced fingers) must be avoided. Building on stable grasps, finger-gaited manipulation is a key component for enabling robust and efficient dexterous operation. In this process, while some fingers maintain contact to

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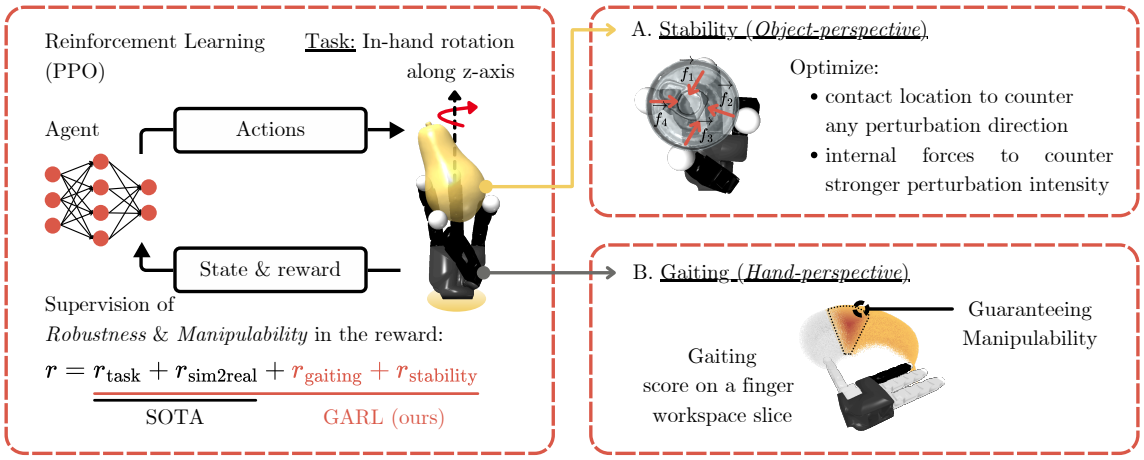


Figure 1: **Overview of the GARL Pipeline.** Our approach augments standard reinforcement learning rewards for *Manipulation* (r_{task}, r_{sim2real}) with analytical metrics derived from robotic grasping theory to improve robustness and manipulability. In (A) grasps are represented in 2D from the top for simplicity. In (B) the workspace of the index finger is represented and splitted in two: one part shows the manipulability metric, ranging from yellow (near singular) to red (high manipulability). The other half of the workspace is shaded in grey for visualization.

preserve wrench balance and others reposition, maintaining manipulability is essential for effective object control.

Existing reward formulations for finger gaiting often produce non-uniform gaiting lengths or poorly conditioned finger motions from a kinematic perspective (Qi et al., 2022). To this end, we propose several finger-gaiting approaches that encourage joint configurations away from singularities while minimizing torque, thereby improving overall system performance and enabling more effective in-hand manipulation. We systematically evaluate several reward formulations for this behavior.

We propose GARL, Grasp-Aware Reinforcement Learning, which integrates both robustness-oriented and manipulability-oriented considerations into the reinforcement learning pipeline of Hora (Qi et al., 2022). The approach builds upon the principles of Rapid Motor Adaptation (Kumar et al., 2021) to enhance simulation-to-reality transfer. We focus on "fingertip in-hand rotation" task on an Allegro hand (Wonik Robotics), i.e. rotating an object about an axis relative to the hand. Our contributions are focused on the privileged teacher policy within the Hora framework, as the capabilities of a distilled student are fundamentally limited by its teacher's performance. Although real-world deployment requires subsequent distillation and sim-to-real transfer, these stages rely on well-documented prior work. By focusing on the teacher, we can rigorously evaluate how our grasp-informed inductive bias improves robustness and manipulability before it is inherited by the student policy.

Our main contributions are:

1. We incorporate desirable quantitative measures of guaranteed object stability, internal-force controllability and manipulability into dedicated rewards to promote robust and dexterous in-hand rotation policies within an RL framework. This enables a comprehensive analysis of the learned policies, characterizing the safety margin against undesired configurations.
2. We extend robustness across the continuous $SO(3)$ space of orientations, rather than a fixed subset, achieving policies with no measurable orientation-dependent performance degradation and improved sample efficiency compared to the baseline. Additionally, our method yields a 25× improvement in rotation reward when generalizing to unseen perturbation intensities.
3. We highlight the role of considering finger-gaiting reward, whose contribution has not yet been clearly characterized from a purely kinematic perspective within an RL framework. We introduce a reward formulation that yields a 18.1% improvement in task performance while reducing average joint torques by 26.7%.
4. We validate these results on objects from the YCB benchmark showing that our approach provides a more reliable training procedure for consistent results.

The remainder of this paper is organized as follows. After reviewing related work (Sec. 2) and analyzing their limitations (Sec. 2.4), we detail how our proposed desirable behaviors in dexterous manipulation are incorporated into the RL-compatible framework in Sec. 3. We then present a comprehensive

set of experiments (Sec. 4) that analyze robustness across perturbation directions and intensities; quantify its sample efficiency; highlight impact of different finger gaiting reward on task performance and torque; and study the generalization of GARL to other objects from YCB (Calli et al., 2015).

2 RELATED WORKS

2.1 Dexterous Manipulation through Analytical Control.

Dexterous in-hand manipulation has been studied for decades. Early methods commonly leverage object and robot kinematic models for motion planning and control in manipulation tasks (Leveroni and Salisbury, 1996; Fearing, 1986; Fan et al., 2017; **?**). These approaches explicitly account for contact constraints and often impose nonholonomic assumptions. As a result, these controllers are highly sensitive to modeling inaccuracies, unmodeled contact dynamics, and simplifying assumptions regarding friction and rigidity, thereby limiting their applicability in unstructured, real-world environments. By contrast, learning-based approaches promise greater adaptability and generality (Yang et al., 2024; Petrenko et al., 2023).

2.2 Dexterous Manipulation through Learning Approaches.

Learning approaches for dexterous manipulation broadly fall into two categories: Imitation Learning (IL) and Reinforcement Learning (RL). Imitation learning is frequently used in two-finger gripper settings (e.g., ACT (Zhao et al., 2023), DexForce (Chen et al., 2025), Octo (Team et al., 2024)) where demonstrations are available. In multi-fingered hands, demonstrations are collected via motion-capture gloves (Wang et al., 2024; Shaw and Pathak, 2024) or directly through RGB / RGB-D camera and via hand pose estimation and hand pose retargeting (Handa et al., 2019; Sivakumar et al., 2022). When the robot morphology closely matches the human one, IL can be sample-efficient and avoid a large domain gap (Antotsiou et al., 2018; Wang et al., 2024; NVIDIA et al., 2025).

However, demonstration collection is time-consuming and requires expensive or elaborate hardware and calibration. Teleoperation also faces latency and haptic feedback limitations, which constrain the difficulty of tasks that can be demonstrated (for example, precise in-hand rotations requiring subtle tactile cues). Morphological mismatch between humans and robotic hands, like the Shadow's DEX-EE, further limits direct transfer of demonstrations, reducing the applicability of IL for fully dexterous in-hand rotation on multi-fingered hands.

Reinforcement Learning with GPU-accelerated simulators, like Isaac Gym (Makoviychuk et al., 2021), makes large-scale training feasible and enables controllers to generalize across objects and tasks (Handa et al., 2024; Petrenko et al., 2023). The primary challenge for RL is simulation-to-reality transfer. But solutions exist and include domain randomization (OpenAI et al., 2019; Handa et al., 2024), system identification (OpenAI et al., 2019), adaptive methods such as Rapid Motor Adaptation (RMA) (Qi et al., 2022; Yang et al., 2024), and the use of sensors/modalities that are realistic in simulation (e.g., low-dimensial binary tactile sensors (Yin et al., 2023), point clouds (Yuan et al., 2023)). Accordingly, we employ a Reinforcement Learning approach (PPO) to enable in-hand rotation using fingertips.

2.3 Robustness to External Forces.

Robustness to varying external forces is critical for in-hand rotation, yet most prior work assumes a static hand orientation relative to gravity (Qi et al., 2022; OpenAI et al., 2019; Ma et al., 2023). In real-world deployment, hands mounted on moving arms experience continuous changes in gravity direction and acceleration. This requires dexterous policies to be robust to external perturbations.

To date, AnyRotate (Yang et al., 2024) is the only work to directly address this via discrete orientation training and task-specific rewards. However, their performance remains highly orientation-dependent, showing a 42% performance drop between "Palm Up" and "Thumb Down" orientations.

We extend this frontier in two ways. 1) Moving beyond discrete to continuous SO(3) hand orientations during training. 2) We evaluate on object masses up to 0.75 kg, a 2.5$\times$ increase over the limit used in prior benchmarks (Qi et al., 2022; Yang et al., 2024).

2.4 Limitations of Baselines in Addressing Grasp Aware Metrics

We build upon well-established theoretical results in grasp analysis to quantify grasp robustness during in-hand manipulation tasks. In Fig. 2, we evaluated a state-of-the-art policy with respect to the grasp-stability metrics defined in Section 3.2. At multiple instances, the grasp-stability metric for the pol-

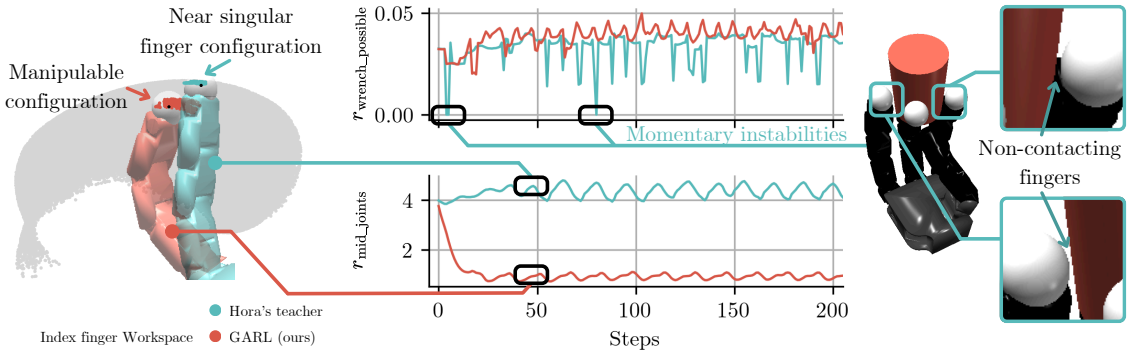


Figure 2: **Comparison of workspace utilization and grasp stability. On the left:** Reached subspace of the index finger during in-hand rotation, highlighting exemplar configurations for the Hora (Qi et al., 2022) baseline (in blue) and GARL (in red). The baseline straight finger corresponds to a high $r_{\text{mid_joints}}$ penalty, while GARL shows a folded finger configuration with enhanced manipulability. **On the right:** Evolution of grasp stability ($r_{\text{wrench_possible}}$) for the two policies from identical initial conditions. Insets highlight momentary instabilities in the baseline where the grasp matrix loses full rank, and in which external perturbations may cause the object to slip or fall.

icy of Hora (Qi et al., 2022) drops to zero, indicating loss of control over at least one degree of freedom. Such events correspond to a temporary loss of force-closure, making the grasp momentarily unstable and prone to slipping or dropping under external perturbations.

In the example illustrated, only two fingertips points are in contact; under a Coulomb friction model this is insufficient to guarantee a full-rank grasp matrix for a general rigid body in 3D, which explains the observed rank loss. Meanwhile, our approach yields non-zero values of the grasp-stability metric for the same initial condition, indicating improved robustness to hand-orientation perturbations. Quantitatively, our approach divides by $2.8\times$ the number of steps with non-full rank grasp matrix (from $6.26\%(\pm 0.7)$ for the baseline to $2.24\%(\pm 0.53)$ with grasp-stability reward).

Along the robustness benefits, Figure 2 (left) illustrates how our GARL framework optimizes workspace utilization compared to the baseline. The point clouds, rendered in blue for the baseline and red for our method, represent the reached subspace of the index finger during in-hand rotation, aggregated over five distinct episodes across four different policies trained. These distributions reveal that the baseline policy consistently operates at the upper boundaries of the finger's workspace. This proximity to kinematic limit makes upward contact motions uncontrollable. The contrast is further highlighted by the exemplar configurations: the baseline (in blue) settles into a nearly straight, singular pose, whereas our approach (in red) favors a folded finger configuration. These configurations are optimized through $r_{\text{mid_joints}}$ which maintains the finger in a more centralized, high-manipulability region of the workspace. This shift from the workspace edge provides the dexterity necessary to sustain the continuous finger-gaiting sequences.

3 GRASP-AWARE APPROACH

Building on the algebraic foundations of grasping, this section presents how our proposed desirable behaviors are incorporated into the RL formulation to achieve robust in-hand manipulation. We extend the typically discrete mathematical criteria into a continuous representation, providing an RL-compatible framework to characterize the safety margin against undesired grasp configurations.

3.1 Notations and Grasp Formalism

We consider a multifingered hand grasping a rigid object. Following the notation in (Prattichizzo and Trinkle, 2008), let the object wrench be denoted by $w \in \mathbb{R}^6$. The interaction between the hand and the object occurs at n_c contact points. While we consider $n_c = 4$ specifically for the fingertip in-hand rotation task, the subsequent grasp map computation naturally scales to a larger number of contacts.

The relationship between the net object wrench w and the vector of local contact wrenches $\lambda \in \mathbb{R}^{kn_c}$ (where k is the number of independent wrench components applied at each contact, e.g. $k = 3$ for the Point Contact with Friction model) is defined by the *Grasp Matrix* $G \in \mathbb{R}^{6\times kn_c}$: $w = G\lambda$. The relationship between the contact wrenches to the required joint torques τ is mediated by the *Hand Jacobian* $J \in \mathbb{R}^{kn_c \times n_q}$: $\tau = J^T\lambda$.

Main objectives are the control of the object wrench w, and the control of the internal forces. Object stability within the hand and manipulability assessments depend on the analysis of the linear subspaces of these two matrices, G and J, which qualify the grasp from both the object's and the hand's perspectives. A comprehensive taxonomy of grasp properties is established based on the conditions on these matrices (Roa and Suárez, 2015).

3.2 Desirable Behaviors

Our method specifically targets the following keypoints.

3.2.1 All Object Wrenches Possible

The achievable object wrenches correspond to the range of G. The condition $\text{Rank}(G) = 6$ ensures that the contacts can geometrically generate a wrench in every direction, allowing the grasp to resist arbitrary external disturbances (such as gravity or arm accelerations). If this condition is not satisfied, the grasp is considered *indeterminate*; there exists at least one direction of external force that the contacts cannot resist, leading to inevitable object loss regardless of actuation effort.

To quantify the safety margin associated with the above algebraic rank condition, our approach considers the Singular Value Decomposition (SVD) of the matrix G. In particular, values of the smallest singular value $\sigma_{\min}(G)$ approaching zero indicate increased vulnerability along specific directions due to impending rank deficiency. Fig. 2 illustrates how $r_{\text{wrench_possible}} = \sigma_{\min}(G)$ evolves through time and specifically, that the baseline (Qi et al., 2022) is null at several points. Such events correspond to momentarily unstable grasps which are prone to slipping or dropping under small external perturbations.

3.2.2 Controllability of all Internal Forces

Wrench intensities in the null space of G are referred to as internal object forces. These contact forces, λ_{int}, generate zero net wrench on the object, but are essential for counteracting undesired external perturbations or maneuvers by maintaining the grasp's tightness. To enable the hand to actively regulate these internal forces via motor torques, the joint configuration q must be chosen so that the internal forces do not reside in the null space of $J(q)^T$.

For computational convenience, the condition for controllability of the internal forces is expressed as

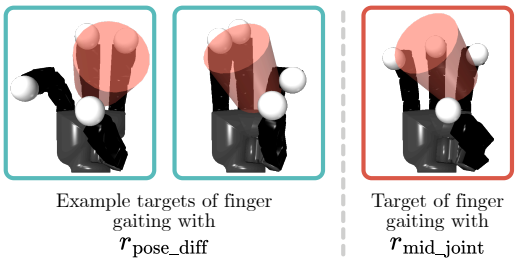


Figure 3: **Target positions used in the finger gaiting reward for $r_{\text{pose_diff}}$ and $r_{\text{mid_joints}}$**

$\text{rank}\begin{pmatrix} G \\ J^T \end{pmatrix} = kn_c$ in our case. We recast this discrete criterion into a continuous metric by maximizing the smallest singular value $\sigma_{\min}\begin{pmatrix} G \\ J^T \end{pmatrix}$, thereby quantifying the margin to rank deficiency.

Hereafter, we denote the resulting grasp rewards as follows:

$$r_{\text{stability}} = \beta_1 r_{\text{wrench_possible}} + \beta_2 r_{\text{intern_forces}} \quad (1)$$

where $r_{\text{wrench_possible}} = \sigma_{\min}(G)$ and $r_{\text{intern_forces}} = \sigma_{\min}\begin{pmatrix} G \\ J^T \end{pmatrix}$.

Remark: Contact locations are required to compute the G and J matrices; however, this requirement is strictly limited to the training phase for reward computation. At inference, when deploying the distilled policy in the real world, neither the reward nor the explicit contact locations need to be computed.

3.2.3 Finger-gaited Manipulation

Configurations near workspace limits or singularities must be avoided, as they constrain the hand's ability to generate controllable velocities. In finger-gaited manipulation, where some fingers maintain contact to preserve wrench balance while others reposition to continue object motion, maintaining manipulability is crucial for effective object control. Without an explicit finger-gaiting reward r_{gaiting}, reinforcement learning agents tend not to develop this behavior, as demonstrated by the experiments in Sec. 4.5. This occurs because the gaiting process involves a long sequence of zero or negative rotation rewards. To overcome this exploration challenge, the reward structure must balance the standard rotation reward with a penalty that accounts for gaiting actions.

The penalty is typically expressed as the distance between the current joint configuration q and a target joint position. However, the effectiveness of this reward is highly sensitive to the choice of target, which

remains a key design challenge in dexterous manipulation. Multiple reward formulations are evaluated for comparison in our study.

- Prior work, such as (Qi et al., 2022), utilizes the initial hand joint configuration as the target, $r_{\text{pose_diff}} = -\|q - q_{\text{init}}\|^2$, where q_{init} is chosen to empirically ensure grasp stability. This target joint position is determined prior to training the policy, using uniform sampling. We argue that this choice could be sub-optimal because the initial configurations might place the hand in positions that are poorly suited for manipulation (such as fingers not contacting the object, unevenly distributed contacts, or fingers pressed too closely together). These configurations can limit gaiting motions to small, constrained movements, which in turn reduces both the speed and smoothness of the object's rotation (Fig. 3).
- To address these limitations, we propose $r_{\text{mid_joints}}$, which uses the middle of each joint's range of motion, $\bar{q}$, as the target configuration:

$$r_{\text{mid_joints}} = -\|q - \bar{q}\|^2 \tag{2}$$

  By employing this metric, $r_{\text{mid_joints}}$ prevents the agent from reaching singular finger configurations or joint limits, thereby preserving manipulability. Note that rather than relying on classical Jacobian-based indices, we formulate manipulability as a kinematic proxy based on distance to joint limits to explicitly optimize workspace utilization during continuous operation.
- Naive quadratic centering of all joints via $r_{\text{mid_joints}}$ may cause the thumb to lose contact with the object. To preserve the thumb's critical oppositional role during finger gaiting, we propose two alternative formulations.
  - The first relaxes the centering constraints on the two basal thumb joints, yielding $r_{\text{mid_joints_relax}}$.
  - The second formulation targets, exclusively for the thumb, its initial joint configuration rather than the joint midpoints, defining $r_{\text{mid_joints_init}}$, while the joint midpoints remain used for the other fingers.
- To account for joint limits and thus to workspace boundary, we propose to consider for further analysis an other alternative reward, $r_{\text{workspace_boundaries}}$, defined by the distance between the distal phalanges and the workspace limit.

The benefits of the aforementioned reward design choices, along with comparisons among them and against a variant without a finger-gaiting reward, are further demonstrated in the experiments presented in Section 4.5.

4 EXPERIMENTS AND ANALYSIS

This section evaluates the effectiveness of incorporating the previously introduced manipulation-aware desired behaviors into a reinforcement learning framework for dexterous manipulation with a four-fingered gripper. The algorithms are trained and assessed on an object rotation task. Before generalizing to objects from the YCB dataset, experiments are first conducted using a cylindrical object with randomized physical properties. The task is designed to replicate complex contact interactions characteristic of finger-gaited manipulation, while accounting for the presence of undesired external perturbations.

4.1 Simulation Setup

We adopt as a baseline the teacher policy from Hora (Qi et al., 2022), trained with privileged information similar to that used in (Yang et al., 2024), due to the availability of open-source code.

The experiments were done in the Isaac Gym simulator (Makoviychuk et al., 2021). We trained our policies using Hora's teacher (Qi et al., 2022) training procedure (largely based on PPO (Schulman et al., 2017)) with $32,192$ parallel environments. Each environment contains an Allegro Hand and a cylinder with randomized physical properties (size, mass, center of mass, and friction coefficient). The hand is controlled at 20Hz, while the physics simulation runs at 120Hz. Each episode lasts a maximum of 400 steps, i.e. 20s of simulated time. All experiments were averaged on 4 seeds for robust results and standard deviation is included inside parenthesis.

4.2 Task

We train a control policy to perform continuous in-hand rotation around the z-axis at 0.5 radian per second. The task is modeled as a Markov Decision Process (MDP) aiming to maximize the expected discounted return.

4.2.1 Observation & Action Space

The policy receives a base observation $o_t \in \mathbb{R}^{121}$ comprising a three-step history of joint positions q and previous actions a, alongside privileged simulation data (object position, size, mass, friction, center of

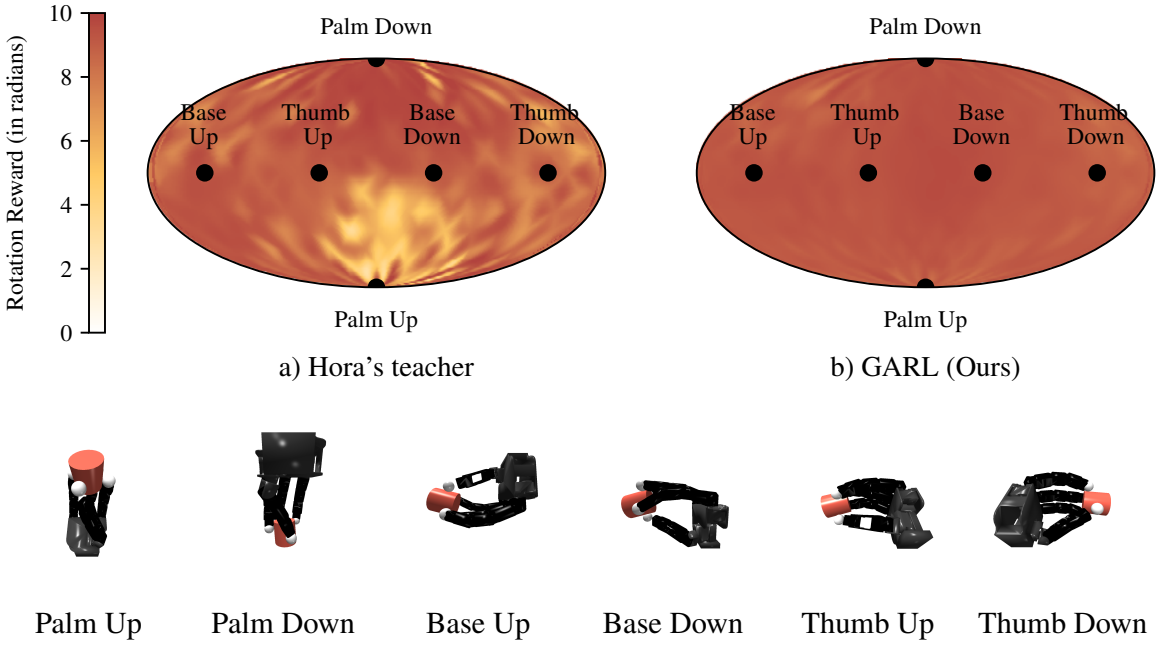


Figure 4: **Robustness to unseen hand orientations.** The planispheres display the mean rotation reward for (a) the Hora's teacher (baseline) and (b) GARL. Black markers indicate the six canonical hand orientations used during training. The baseline fails to generalize to many orientations, producing large regions of poor performance, whereas our method yields a substantially more uniform performance distribution over the sphere.

mass, hand orientation and contact forces). To ensure orientation invariance, coordinates are expressed in the hand-local frame (except for the hand orientation). The action space $a_t \in \mathbb{R}^{16}$ consists of relative joint-position targets, which a 20Hz PD controller converts to motor torques.

4.2.2 Reward Function

The reward r encourages stable rotation while penalizing energy expenditure to facilitate sim-to-real transfer:

$$r = r_{\text{task}} + r_{\text{sim2real}} + r_{\text{stability}} + r_{\text{gaiting}} \tag{3}$$

where $r_{\text{task}} = \alpha_0 r_{\text{rot}} + \alpha_1 r_{\text{linvel}}$, $r_{\text{sim2real}} = \alpha_2 r_{\text{work}} + \alpha_3 r_{\text{torque}}$. Here, r_{rot} corresponds to the clipped angular velocity ω about the target axis $\hat{k}$. Penalties are imposed on the object's linear velocity (v) through r_{linvel}, on mechanical work $\tau^T \dot{q}$ via r_{work}, and on torque magnitude $\|\tau\|^2$ via r_{torque}. Our primary contributions, $r_{\text{stability}}$ and r_{gaiting}, are described in detail in Section 3.2.

4.2.3 Evaluation Metrics

We employ three metrics to evaluate the effectiveness of both our method and the baseline:

- Time-To-Fall (TTF) is the episode length until the object is dropped or the policy reaches the maximum number of steps (i.e. 400), normalized to $[0, 1]$;
- Rotation Reward (RotR) is the clipped mean rotation achieved per episode (in radians);
- Sample Efficiency (SE) is the number of training steps required to reach 95% of the asymptotic RotR.

4.3 Robustness Analysis under Varying Directions and Magnitudes of External Perturbations

Generalization to unseen perturbation directions and magnitudes applied to the object is investigated below.

4.3.1 Robustness to Unseen Directions

For comparison purposes, training of the baseline (i.e., without Grasp Stability) from Hora (Qi et al., 2022)'s teacher and GARL (ours) considers only six canonical object orientations, inspired by (Yang et al., 2024): palm up, palm down, base up, base down, thumb up, and thumb down (Fig. 4). Their policies

were then evaluated for 300000 episodes with uniformly randomized hand orientations in $SO(3)$.

Policy evaluation under uniformly randomized orientations in $SO(3)$ reveals that the baseline exhibits large regions of poor generalization performance, whereas our method achieves a substantially more uniform performance distribution. The stability term encourages contact strategies that transfer more effectively across orientations. Quantitatively, incorporating the grasp-stability reward yields a 15.1% improvement in rotation reward compared to the baseline.

4.3.2 Robustness to Unseen Intensities

Variations in perturbation magnitude may arise from robot arm accelerations, contacts with the environment, or changes in the object's mass. Generalization is evaluated over magnitudes ranging from 0 to $3g$, with g representing standard terrestrial gravity. Policies were trained on two intensity ranges: (A) $[0,1]$g and (B) $[0,3]$g, every time with "palm-up" orientation only. For each test intensity we ran 10000 episodes and report the mean Rotation Reward in Fig. 5. Across all training and evaluation intensities, our method outperforms the baseline. For example, when trained on $[0,1]$g and evaluated at the strongest test condition (3g), the baseline is unable to hold the object (RotR: 0.1 (±0.1) and TTF: 0.06 (±0.01)), while our approach does succeed to hold and rotate the object (RotR: 2.5 (±0.3), TTF: 0.44 (±0.08)), an improvement of 25× on rotation reward. The baseline trained on $[0,1]$g degrades sharply beyond its training range (loss of rotation reward immediately after 1g), while our method shows a smoother degradation, indicating better extrapolation to larger forces. When training with the wider intensity distribution $[0,3]$g, the relative improvement in rotation reward shrinks to 68.6% (baseline: 3.1 ±2.0, GARL: 5.3 ±0.8), and share similar degradation decay.

Qualitatively, the combination of $r_{\text{wrench_possible}}$ and $r_{\text{intern_forces}}$ in GARL promotes frequent and timely short finger-gaiting motions. This strategy maximizes the number of contact points (crucial under large external forces for maintaining stability) while ensuring controllable internal forces to enhance the object's squeezing effect.

4.4 Analysis on Sample Efficiency

While domain randomization and an appropriate observation space can yield robust policies, vanilla PPO typically requires large amounts of experience for broad generalization. A critical question is how many training steps are necessary to achieve robustness to external perturbations. We show that our manipulation-aware reward also provides a strong inductive prior for robustness, reducing the required training steps substantially.

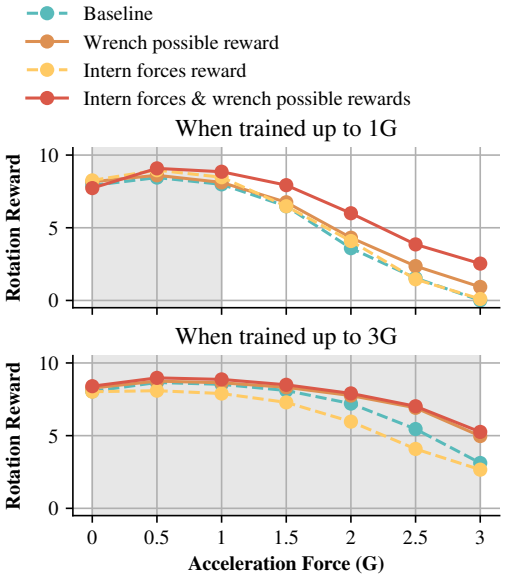


Figure 5: **Robustness to unseen intensities.** Evaluation across 0-3g, our method outperforms the baseline at all magnitudes — notably a 25× relative improvement in Rotation Reward at 3g for the policy trained on $[0,1]$g. The grey background represents the training randomization of acceleration forces.

Table 1: **Sample efficiency for different numbers of directions and intensity ranges of gravity used in training.**

Ext. Perturb.		# steps (×100M) ↓		Ratio ↑
# Dir.	Intensity (in G)	Hora's Teacher	GARL (Ours)	
1	[0, 1]	4.4 (±0.4)	**1.4** (±0.4)	3.1×
6	[0, 1]	17.5 (±4.2)	**1.4** (±0.4)	12.5×
SO(3)	[0, 1]	10.5 (±4.8)	**2.1** (±1.5)	5.0×
1	[0, 1]	4.4 (±0.4)	**1.4** (±0.4)	3.1×
1	[0, 2]	11.1 (±2.5)	**5.0** (±2.9)	2.2×
1	[0, 3]	27.5 (±4.5)	**11.5** (±2.8)	2.4×

Table 1 gather the sample efficiency of policies trained for 3 billion steps, and averaged across four seeds. In the six-direction training setup the baseline required 1.75B environment steps to reach the target rotation performance, whereas our method reached the same target in about 140M steps (an improvement of 12.5× in sample efficiency). Moreover, increasing the number of randomized perturbation directions (from 1 to 6 and up to full $SO(3)$ coverage) did not significantly increase the required training steps for

our method, demonstrating that it attains extensive directional robustness with minimal additional training effort. We also observe that, as the range of perturbation intensities increases, both methods scale linearly; however, GARL demonstrates a $4.7\times$ improvement in sample efficiency in average.

These results demonstrate that the grasp-stability reward not only enhances asymptotic robustness but also significantly improves sample efficiency for learning orientation-robust manipulation. From the training perspective, our grasp stability reward acts as a strong inductive bias. In the early stages of training, when the policy moves randomly, the $r_{\text{stability}}$ term provides immediate, dense feedback, guiding the fingers toward stable grasps, therefore increasing significantly the Time-To-Fall (TTF). As training progresses, it penalizes "precarious" grasps (those that might technically hold the object but rely on singular, unstable contact arrangements), thereby favoring the emergence of behaviors that inherently satisfy the criteria of controlling wrenches and internal forces.

4.5 Finger-Gaiting Manipulation Analysis

In this section, we assess the impact of our previously proposed desired criteria, focusing on how different finger-gaiting reward formulations influence policy behavior and task performance (Table 2). These experiments were undergone with the "palm-up" orientation.

4.5.1 On the Necessity of the Finger-Gaiting Reward

Sustained in-hand rotation requires finger gaiting, which does not naturally emerge within the RL pipeline without explicit incentivization. As shown by the baseline $\varnothing\, r_{\text{gaiting}}$ results in Table 2, the Rotation Reward plateaus at approximately 5.27 in the absence of a dedicated finger-gaiting reward. In this case, the learned policy converges to a strategy in which the fingers gradually reach their joint limits and subsequently hold the object static.

4.5.2 Enhancing Manipulability

The choice of reward formulation significantly impacts the agent's ability to maintain manipulability and achieve sustained rotation. As discussed in Sec. 3.2.3, the issues of $r_{\text{pose_diff}}$ suggested in (Qi et al., 2022) lead to suboptimal finger placements that penalize long-term rotation performance. Among the alternatives, $r_{\text{workspace_boundaries}}$ provides a sparse signal relative to direct joint penalization, resulting in diminished performance. In contrast, our proposed $r_{\text{mid_joints}}$ formulation and its variants demonstrate superior effectiveness in terms of Rotation Reward, as they more effectively penalize configurations near joint limits. This design yields a 18.1% improvement in Rotation Reward over $r_{\text{pose_diff}}$, while simultaneously reducing the average joint torque demand by 26.7%. As a result, the learned policies favor sequences of grasps with enhanced manipulability, where small joint motions produce large contact displacements, thereby lowering torque requirements.

Table 2: **Impact of different finger gaiting rewards.** Ran on 4 seeds, without any robustness reward as introduced in 3. $r_{\text{mid_joints}}$ brings a 18.1% improvement of rotation reward over $r_{\text{pose_diff}}$.

r_{gaiting}	RotR ↑	TTF ↑	Torque ↓
$\varnothing\ r_{\text{gaiting}}$	5.3 (±0.7)	**.96** (±0.01)	2.0 (±0.3)
$r_{\text{pose_diff}}$	7.2 (±0.2)	**.96** (±0.01)	1.9 (±0.3)
$r_{\text{mid_joints}}$	8.4 (±0.4)	.94 (±0.00)	1.7 (±0.2)
$r_{\text{mid_joints_init}}$	**8.5** (±0.0)	.90 (±0.01)	**1.5** (±0.1)
$r_{\text{mid_joints_relax}}$	**8.5** (±0.0)	.90 (±0.00)	**1.5** (±0.1)
$r_{\text{workspace_boundaries}}$	7.4 (±0.3)	.96 (±0.01)	2.1 (±0.3)

4.6 Object Generalization

To assess object generalization of our method, we independently trained and evaluated GARL in the palm-up orientation on different objects from the "Food" subset of the YCB dataset (Calli et al., 2015). As detailed in Table 3, GARL consistently maintains a high level of performance regardless of object complexity. While the baseline suffers substantial degradation on the cube (3.1 RotR) and tomato soup can (3.4 RotR), GARL maintains an average RotR above 5.9 across all test cases. Although a marginal performance trade-off is observed on the mustard bottle case (-9% RotR), GARL yields substantial improvements on two of the six evaluated objects (e.g. $+103\%$ for the tomato soup can) and achieves comparable or improved metrics on the remainder. The baseline's inconsistencies are primarily driven by specific random seeds failing to maintain stable grasps throughout the manipulation, resulting in high metric variance. By guiding the agent's exploration toward stable grasps early in the learning process, GARL circumvents this issue, providing a significantly more reliable and consistent training procedure.

Table 3: **Performance Across YCB Food Objects**

Object	Method	RotR ↑	TTF ↑
Cylinder	Hora	7.1 (±0.3)	**.96** (±0.01)
	GARL	**8.4** (±0.4)	.94 (±0.00)
Cube	Hora	3.1 (±2.2)	.42 (±0.27)
	GARL	**5.9** (±1.1)	**.78** (±0.06)
Tomato Soup Can (YCB)	Hora	3.4 (±4.2)	.43 (±0.42)
	GARL	**6.9** (±0.4)	**.85** (±0.00)
Mustard Bottle (YCB)	Hora	**7.6** (±0.1)	.89 (±0.02)
	GARL	6.9 (±0.4)	**.93** (±0.02)
Strawberry (YCB)	Hora	7.3 (±0.2)	**.79** (±0.01)
	GARL	**7.4** (±0.4)	**.79** (±0.03)
Pear (YCB)	Hora	**7.5** (±1.0)	**.83** (±0.09)
	GARL	7.3 (±0.5)	.82 (±0.05)

5 CONCLUSION

Our approach incorporates both object-related and hand-related considerations within a reinforcement learning framework (Proximal Policy Optimization in our case) to achieve desired behaviors for dexterous manipulation tasks. Specifically, the method targets key aspects of multifingered grasps through quantitative measures of guaranteed stability, controllability, and manipulability. This enables a comprehensive analysis of the learned policies, characterizing the safety margin against undesired grasp configurations.

Collectively, these design considerations allow the control policy to perform dexterous object manipulation while maintaining grasp stability under external perturbations. The hybrid robotic grasping-guided RL approach offers several advantages: (i) it drives the discovery of contact locations that substantially improve robustness to unseen external perturbations across both magnitude and direction, (ii) it provides an inductive bias that accelerates learning by up to twelvefold. Furthermore, the learned finger-gaiting policies enhance task performance by promoting grasp sequences with greater manipulability, in which small joint motions produce large contact displacements, thereby lowering torque demands.

We validate these results on objects from the YCB benchmark showing that our approach provides a more reliable training procedure for consistent results.

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